

GSIF RESEARCH NETWORK

Global Sustainability Impact Foundation

PRODUCT REQUIREMENTS DOCUMENT (PRD)

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1. Introduction & Product Overview

GSIF Research Network is a dedicated digital platform built for the Global Sustainability Impact Foundation. The platform connects individual researchers and research institutions around the world, enabling them to collaborate, publish research, discover events, and build meaningful professional relationships — all centered on sustainability and impact-driven science.

1.1 Product Vision

To become the leading global research collaboration hub where sustainability researchers and institutions discover each other, share knowledge, and collectively drive meaningful change across the world's most pressing environmental challenges.

- Create a verified sustainability research ecosystem.
- Enable structured research publishing (PDF-based).
- Support collaboration between researchers and institutes.
- Provide global research visibility.
- Facilitate sustainability-focused events and networking.

2. Functional Modules Overview

- Landing Page
- User Registration (Individual & Institute)
- Login & Forgot Password (OTP Based)
- Profile Completion Workflow
- Admin Panel (Approval & Moderation)
- Dashboard & Feed System
- Research Publishing & Library
- Events Management System
- Chat & Messaging System
- Notification System

3. User Types

- Individual Researcher — A standalone researcher, student, educator, or professional who registers independently and completes a 6-step profile.
- Research Institute — An organization (university, NGO, company, government body) whose registration must be approved by the GSIF admin before gaining platform access. Managed by a designated representative.
- GSIF Admin — Manages approvals, content moderation, events, and platform-wide communications via Admin Panel.

4. Technology Stack (Reference)

- Frontend: HTML, CSS, Tailwind CSS, JavaScript, React.js
- Backend: PHP (CodeIgniter)
- Database: MySQL
- File Storage:
- Server: Hostinger
- Real-time: Pusher
- Email:

5. Feature Requirements

1. User Management & Roles

1.1 User Types

- **Individual Researcher**
- **Research Institute**
Each institute account is managed by an authorized representative.

1.2 Role-Based Access Control

- Features and permissions are assigned based on user role.
- Admin users have verification, approval, and moderation privileges.

2. Registration & Onboarding

2.1 Individual Researcher Registration

- Users can register using basic personal details.
 - After registration:
 - A **Welcome Screen** is displayed.
 - Users are prompted to complete their profile.
 - Profile completion is mandatory and consists of a **6-step guided process**.
 - Dashboard access is granted only after successful profile completion.
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2.2 Research Institute Registration

- Institute registration is initially set to **Under Review**.
- **Admin approval** is required before activation.
- Upon approval:
 - An approval notification email is sent to the registered email address.
 - Account status is updated to **Approved**.
- Approved institutes must complete a **5-step profile setup**.
- Dashboard access is provided after profile completion.

3. Profile Management

3.1 Individual Researcher Profile

- Personal and professional information
- Research role and academic level
- Sustainable Development Goals (SDGs) selection
- Experience, skills, tools, and previous publications
- Collaboration interests

- Professional profile links (LinkedIn, ResearchGate, ORCID, personal website)

3.2 Research Institute Profile

- Institution type
- Location details (country, state, city)
- Primary research focus areas
- Platform objectives
- External professional profile links

3.3 Profile Visibility

- Users can view other researchers' profiles.
- Users can visit and explore research institute profiles.

4. Research Publishing & Library

4.1 Research Upload

- Users can publish research in **PDF format**.
- Required metadata includes:
 - Title
 - Abstract or summary
 - Relevant tags or domains

4.2 Research Preview

- A preview of each research publication appears in the home feed.
- Selecting “**View Full**” redirects the user to the **Research Library** page.

4.3 Research Library

- Centralized repository for all published research.
- Designed to support future search and filtering capabilities.

5. Home Feed & Content Posts

5.1 Supported Content Types

Users can create and publish:

- Articles
- Polls
- Images
- Research previews

5.2 Feed Behavior

- The home page displays a continuous scrolling feed.
- Content includes user posts and research previews.
- Newer and relevant content is prioritized.

6. Events Management & Notifications

6.1 Event Types

- **GSIF-organized events**
- **Externally organized events** (subject to approval)

6.2 Event Approval Workflow

- External organizers must submit an event request for approval.
- Admin reviews and approves or rejects the request.
- Approved events are published on the platform.
- Event notifications are sent to all users.

6.3 Event Notifications

- Users receive notifications for:
 - Upcoming events
 - Event updates
 - Platform announcements

7. Communication & Collaboration

7.1 Chat System

- One-to-one messaging between researchers.
- Secure in-platform communication.
- Supports real-time or near real-time messaging.

7.2 User Connections

- Users can connect with other researchers or institutes.
- Connections enable easier collaboration and communication.

8. Admin Panel Features

8.1 Institute Verification

- Review and approve or reject institute registrations.
- Manage institute account status.

8.2 Event Moderation

- Review and manage event permission requests.
- Control event visibility and publishing.

8.3 Platform Governance

- Monitor user activities.
- Moderate platform content (future scope).

9. Dashboard Features

9.1 Individual Dashboard

- Profile overview
- Research upload and management
- Posts and activity summary
- Notifications
- Chat access

9.2 Institute Dashboard

- Institute profile management
 - Research publishing
 - Event creation (after approval)
 - Collaboration tracking
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10. Notification System

- Registration and approval notifications
- Event alerts
- Platform updates
- User activity notifications

6. Detailed Epics & User Stories

EPIC 1: LANDING PAGE

The public-facing marketing page. No login required. Introduces GSIF Research Network, its mission, and guides visitors to register as either an Individual or Institute.

Sections Included

- Navigation Bar — GSIF Research Network logo (left), Login button (right)
- Hero Section — Main heading, sub-heading, and two CTA buttons: Join as Individual & Join as Institute
- About GSIF Research Network
- Our Impact — Where Research Drives Change
- Our Mission & Our Vision
- Who Can Join? — Individual Researcher / Institute Researcher
- OUR RESEARCH ECOSYSTEM — 6 feature highlights
- How Research Moves Forward — 4-step process
- Built for Every Field — 4 sustainability domains
- Footer

User Stories

US-1.1

As a **website visitor**, I want to **understand what GSIF Research Network is from the hero section**, so that I can quickly decide if this platform is relevant to me

Acceptance Criteria:

- ✓ GSIF Research Network branding is visible in the top-left navbar
- ✓ A Login button appears in the top-right of the navbar
- ✓ Hero section displays a clear main heading and sub-heading
- ✓ Two CTA buttons are present: Join as Individual and Join as Institute
- ✓ Both CTA buttons navigate to their respective registration pages

US-1.2

As a **website visitor**, I want to **read about GSIF's mission, vision, and impact**, so that I can understand the foundation's purpose before joining

Acceptance Criteria:

- ✓ About section clearly explains GSIF Research Network
- ✓ Impact section communicates research-driven change
- ✓ Mission and Vision are displayed as separate statements

US-1.3

As a **website visitor**, I want to **see which user types can join the platform**, so that I know which registration path applies to me

Acceptance Criteria:

- ✓ Who Can Join section shows Individual Researcher and Institute Researcher
- ✓ Descriptions are distinct for each user type

US-1.4

As a **website visitor**, I want to **explore the research ecosystem features**, so that I understand what the platform offers before registering

Acceptance Criteria:

- ✓ Research Publishing Platform is highlighted
- ✓ Research Discovery & Library is mentioned
- ✓ Researcher & Institute Profiles are showcased
- ✓ Collaboration & Networking is explained
- ✓ Global Research Visibility is mentioned
- ✓ Smart Notifications & Updates are described

US-1.5

As a **website visitor**, I want to **understand the research journey in steps**, so that I know what to expect after joining

Acceptance Criteria:

- ✓ 4-step process is shown: Create Profile, Discover Opportunities, Connect & Collaborate, Share Insights
- ✓ Steps are visually sequential and easy to follow

US-1.6

As a **website visitor**, I want to **see which research fields the platform covers**, so that I know if my domain is supported

Acceptance Criteria:

- ✓ Four fields displayed: Sustainable Agriculture & Food System, Water Resources and Conservation, Climate Change & Environment, Circular Economy and Waste Management

EPIC 2: USER REGISTRATION

Two separate registration flows — one for Individual Researchers and one for Research Institutes. Each collects different data relevant to the user type.

2A — Individual Researcher Registration

- Full Name
- Email Address
- Password & Confirm Password
- Address: Country, City, State, Pin Code

2B — Research Institute Registration

- Institute Details: Institute Name, Email, Contact Number, Website Link
- Representative Details: Representative Name, Role, Password, Confirm Password

User Stories

US-2.1

As a **individual researcher**, I want to **register with my personal details on a dedicated form**, so that I can create my account and start using the platform

Acceptance Criteria:

- ✓ Registration form collects: Full Name, Email, Password, Confirm Password
- ✓ Address fields: Country, City, State, Pin Code
- ✓ Password and Confirm Password must match
- ✓ Email format is validated
- ✓ Duplicate email shows a clear error message
- ✓ On success, user is redirected to the Welcome Screen

US-2.2

As a **research institute representative**, I want to **register my institution with both institute and representative details**, so that my institution can apply for a place on the platform

Acceptance Criteria:

- ✓ Form collects institute details: Name, Email, Contact, Website
- ✓ Form collects representative details: Name, Role, Password, Confirm Password
- ✓ All mandatory fields validated before submission
- ✓ On success, user sees the Under Review screen
- ✓ No dashboard access is granted before admin approval

US-2.

As a **any registering user**, I want to **see clear validation errors if I fill the form incorrectly**, so that I can correct my input and successfully submit

Acceptance Criteria:

- ✓ Required field errors shown inline
- ✓ Password mismatch error shown clearly
- ✓ Email format error is shown
- ✓ Form does not submit with invalid data

EPIC 3: LOGIN & PASSWORD RECOVERY

Authentication system covering login and a 3-step OTP-based password reset flow.

Login

- Fields: Email Address, Password
- Redirects to dashboard or profile completion based on user state

Forgot Password Flow

- Step 1: User enters registered email address
- Step 2: OTP is sent to email; user enters OTP to verify

- Step 3: User sets a new password

User Stories

US-3.1

As a **registered user**, I want to **log in with my email and password**, so that I can access my dashboard and platform features

Acceptance Criteria:

- ✓ Login page has Email and Password fields
- ✓ Invalid credentials show an error message
- ✓ Successful login redirects to dashboard (or profile completion if incomplete)
- ✓ Forgot Password link is visible on login page

US-3.2

As a **user who forgot my password**, I want to **request a password reset via my email**, so that I can regain access to my account without contacting support

Acceptance Criteria:

- ✓ User can navigate to Forgot Password from login page
- ✓ User enters registered email and requests OTP
- ✓ OTP is delivered to the registered email
- ✓ OTP input screen accepts 6-digit code
- ✓ Expired or invalid OTP shows an appropriate error
- ✓ After OTP verification, user can set a new password
- ✓ New password must be confirmed; mismatch shows error
- ✓ On success, user is redirected to login with a success message

EPIC 4: COMPLETE YOUR PROFILE (ONBOARDING STEPS)

Post-registration onboarding. Individual users complete 6 steps; Institute users complete 5 steps (only after admin approval).

4A — Individual Researcher Profile Steps

- Step 1: Which best describes you? (Student Researcher, Educator, Professional)
- Step 2: Which SDGs do you work on? (All 17 UN SDGs shown)
- Step 3: What is your current research level? (Undergraduate, Masters Student, PhD Candidate, Postdoctoral Researcher)
- Step 4: Experience & Expertise — Years of experience, Skills/Tools, Previous Publications
- Step 5: Which types of collaborations interest you? (Joint Research, Publication, Workshops & Seminars, Mentorship, Grant Applications, Gain Visibility)
- Step 6: Link your professional profiles (LinkedIn, ResearchGate, ORCID, Personal Website)

4B — Research Institute Profile Steps

Note: Institute must receive admin approval and approval email before accessing profile completion.

- Step 1: Tell us about your institution (University, Research Institute, NGO, Company, Government)
- Step 2: Where is your institution based? (Country, State, City)
- Step 3: Primary research focus areas — predefined list + custom user additions (Oceanography, Water & Waste, Biodiversity, Climate Change, Renewable Energy, Sustainable Agriculture, Urban Planning, Carbon Capture, and more)
- Step 4: What would you like to achieve? (Share & publish research, Increase visibility, Connect with researchers/institutions, Find collaborators, Host sustainability initiatives) + user-defined additions
- Step 5: Link your professional profiles (LinkedIn, ResearchGate, ORCID, Personal Website)

User Stories

US-4.1

As a **newly registered individual researcher**, I want to **see a Welcome Screen after registration with a prompt to complete my profile**, so that I **understand the next step and feel guided**

Acceptance Criteria:

- ✓ Welcome Screen is shown immediately after individual registration
- ✓ A Complete Profile button is clearly visible
- ✓ Clicking the button begins Step 1 of 6

US-4.2

As a **individual researcher**, I want to **complete my profile in 6 guided steps**, so that my profile is rich with information for collaboration discovery

Acceptance Criteria:

- ✓ Progress indicator shows current step (e.g., Step 2 of 6)
- ✓ User can go back to previous steps
- ✓ All 6 steps are completable in sequence
- ✓ On Step 6 completion, a Profile Complete popup/modal appears
- ✓ After dismissing the popup, user lands on their dashboard

US-4.3

As a **institute representative**, I want to **see an Under Review screen after registration**, so that I know my application is pending and I understand the process

Acceptance Criteria:

- ✓ Under Review screen is shown immediately after institute registration
- ✓ Screen clearly explains that admin approval is required
- ✓ No dashboard or profile completion is accessible at this stage

US-4.4

As a **institute representative**, I want to **receive an approval email and see an Approved status on screen**, so that I know when I can proceed to complete my profile

Acceptance Criteria:

- ✓ Admin approval triggers an email to the institute's registered email
- ✓ The institute user's screen updates to show Approved status
- ✓ A prompt/button to begin profile completion becomes available

US-4.5

As a **approved institute representative**, I want to **complete my institution profile in 5 guided steps**, so that my institution is properly set up on the platform

Acceptance Criteria:

- ✓ Progress indicator shows Step X of 5
- ✓ Step 3 allows adding custom research focus areas beyond the predefined list
- ✓ Step 4 allows adding custom achievement goals beyond the predefined list
- ✓ On completion, user gains access to the institute dashboard

EPIC 5: ADMIN PANEL IN PROGRESS

A separate interface for GSIF administrators to manage the platform — approving institutes, moderating content, managing events, and overseeing the research ecosystem.

Key Responsibilities of Admin

- Review and approve/reject institute registration applications
- Send approval/rejection emails to institutes
- Manage event permission requests from users
- Moderate content (posts, research, comments)
- Manage user accounts (suspend, activate)
- View platform analytics and usage statistics

User Stories

US-5.1 IN PROGRESS

As a **GSIF admin**, I want to **view a list of all pending institute registration applications**, so that I can review and process them in a timely manner

Acceptance Criteria:

- ✓ Admin dashboard shows a queue of pending institute applications
- ✓ Each application shows: Institute Name, Representative Name, Email, Submission Date
- ✓ Admin can click to view full registration details

US-5.2 IN PROGRESS

As a **GSIF admin**, I want to **approve or reject an institute application with a note**, so that the institute is notified and can proceed or reapply

Acceptance Criteria:

- ✓ Approve button triggers approval email to institute's registered email
- ✓ Reject button allows admin to add a rejection reason before sending

- ✓ Approved institute's status updates to Approved in the system
- ✓ Rejected institute's status updates to Rejected with reason stored

US-5.3

As a **GSIF admin**, I want to **review and approve or deny event hosting requests from users**, so that only vetted events are published on the platform

Acceptance Criteria:

- ✓ Event requests appear in admin panel with requester details
- ✓ Admin can approve, deny, or request more information
- ✓ On approval, event notification is sent to all relevant users
- ✓ Denied request sends a notification to the requester

US-5.4

As a **GSIF admin**, I want to **manage user accounts including suspension and reactivation**, so that I can maintain platform safety and quality

Acceptance Criteria:

- ✓ Admin can search for users by name or email
- ✓ Admin can suspend an account with a reason
- ✓ Admin can reactivate a suspended account
- ✓ Suspended users see an account suspended message on login

US-5.5

As a **GSIF admin**, I want to **view platform usage analytics**, so that I can report on platform growth and activity to GSIF leadership

Acceptance Criteria:

- ✓ Dashboard shows total registered users (individual + institute)
- ✓ Shows total research papers published
- ✓ Shows total events hosted
- ✓ Shows new registrations over time (chart or table)

EPIC 6: DASHBOARD IN PROGRESS

The personalized home screen for logged-in users. Provides quick access to all platform features, shows activity summaries, and surfaces relevant content.

Dashboard Elements (Both User Types)

- Welcome message with user name and profile completeness indicator
- Quick stats — Research published, connections, events attended
- Navigation to: Home Feed, Research Library, Events, Chat, Profile
- Recommended researchers/institutes to connect with
- Recent notifications

User Stories

US-6.1

As a **logged-in researcher**, I want to **see a personalized dashboard after completing my profile**, so that I have a central hub to access all platform features

Acceptance Criteria:

- ✓ Dashboard shows the user's name and a welcome message
- ✓ Navigation links to: Home Feed, Library, Events, Chat, My Profile
- ✓ Quick stats section shows key numbers relevant to the user
- ✓ Incomplete profile shows a banner prompting completion

US-6.2

As a **logged-in researcher**, I want to **see a summary of my recent activity on the dashboard**, so that I can quickly catch up on what's happened since my last visit

Acceptance Criteria:

- ✓ Recent notifications (up to 5) shown on dashboard
- ✓ Unread message count shown with link to Chat
- ✓ Latest research or posts from my network shown

US-6.3

As a **logged-in researcher**, I want to **discover recommended researchers or institutes on my dashboard**, so that I can expand my network without actively searching

Acceptance Criteria:

- ✓ Recommendations based on shared SDGs, research level, or focus areas
- ✓ Each recommendation shows name, type, and a Connect/View Profile button
- ✓ At least 3 recommendations shown with an option to see more

EPIC 7: RESEARCH PUBLISHING & LIBRARY PENDING

Users can publish research papers in PDF format. Research appears as previews in the Home Feed and in full on the Research Library page.

Key Flows

- Researcher uploads a PDF with title, abstract, keywords, domain/SDG tags
- Research preview card appears in the Home Feed scroll
- Clicking View Full navigates the user to the Research Library page with the complete entry
- Research Library is searchable and filterable

User Stories

US-7.1

As a **researcher**, I want to **upload and publish a research paper in PDF format**, so that my research is discoverable by the GSIF community

Acceptance Criteria:

- ✓ Upload form accepts PDF files only
- ✓ Required fields: Title, Abstract, Keywords, Domain/SDG tags
- ✓ File size limit is clearly communicated
- ✓ Published research appears in the Home Feed as a preview card
- ✓ Research is also listed in the Research Library

US-7.2

As a **platform user**, I want to **see research preview cards in the Home Feed**, so that I can discover new research while browsing my feed

Acceptance Criteria:

- ✓ Preview card shows: title, author, abstract snippet, domain tags, date
- ✓ View Full button navigates to the Research Library detail page
- ✓ Research cards are intermixed with social posts in the feed

US-7.3

As a **platform user**, I want to **search and filter research in the Library page**, so that I can find research relevant to my work efficiently

Acceptance Criteria:

- ✓ Library page has a search bar for title/keyword search
- ✓ Filters available: Domain, SDG, Research Level, Date Published
- ✓ Results paginated or infinite-scrolled
- ✓ Each result links to a full research detail/download page

US-7.4

As a **platform user**, I want to **download or view a full research paper**, so that I can read and use the research in my work

Acceptance Criteria:

- ✓ Research detail page shows full abstract, author info, tags, date
- ✓ Download PDF button available
- ✓ In-browser PDF preview available if possible
- ✓ Author's profile link shown on research detail page

EPIC 8: HOME FEED & SOCIAL POSTS PENDING

The scrollable home feed aggregates social posts and research previews. Users can create three types of posts: Articles, Polls, and Images.

Post Types

- Article — Long-form text post with optional formatting
- Poll — Question with multiple choice options; shows live results
- Image — Single or multiple image upload with a caption

User Stories

US-8.1

As a **logged-in user**, I want to **create and publish an article post on the platform**, so that I can share insights and knowledge with the community

Acceptance Criteria:

- ✓ Post creation modal/page has a type selector: Article, Poll, Image
- ✓ Article post allows a title and rich text body
- ✓ Published article appears in the Home Feed immediately
- ✓ Author name, profile picture, and timestamp shown on the post

US-8.2

As a **logged-in user**, I want to **create a poll and see live voting results**, so that I can gather community opinions on research topics

Acceptance Criteria:

- ✓ Poll creator allows a question and 2-5 answer options
- ✓ Users can vote on a poll once
- ✓ Poll shows current result percentages after voting
- ✓ Poll expiry date can optionally be set

US-8.3

As a **logged-in user**, I want to **upload images with a caption as a post**, so that I can share visual content like posters, infographics, and fieldwork photos

Acceptance Criteria:

- ✓ Image post supports single
- ✓ Caption/description field is available
- ✓ Images are displayed in the feed

US-8.4

As a **logged-in user**, I want to **scroll through a combined feed of posts and research previews on the home page**, so that I can stay updated on both community activity and new research in one place

Acceptance Criteria:

- ✓ Home Feed shows a mix of social posts and research preview cards
- ✓ Feed is ordered chronologically (newest first) with option to switch to Trending
- ✓ Infinite scroll or pagination for loading more content
- ✓ Each post shows author, type badge, content preview, and engagement actions